

Lunar-to-Orbit Neutrino Transit Network: A Critical Conceptual Assessment of Multi-Station Neutrino-Wall Earth-Interior Monitoring and Its Hypothesized Application to Earthquake Detection

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Manuscript version: v0.2 (revises v0.1 by replacing the originally proposed neutrino–neutrino 'collision' mechanism with a multi-station neutrino-transit capture network, per the author's refinement; companion piece to the author's LX-RSRS preprint), September 2026. This note follows the same evidence-grading convention used in LX-RSRS (A: observational evidence; B: established mathematical/physical models; C: demonstrated engineering/experimental technology; D: this paper's own derivation from established parameters; E: untested hypothesis with no known supporting mechanism).

Abstract

This note assesses a revised proposal for a lunar-to-orbit neutrino monitoring architecture and its hypothesized application to earthquake detection and prediction. In this revision, a neutrino source based on the Moon emits a beam that passes through the Earth; multiple receiving 'neutrino wall' stations, placed at various positions in near-Earth orbit (not only on the far side), record arrival timing, flux attenuation, and any anomalous sub-threshold events. The aggregated multi-station data set is intended to support real-time anomaly monitoring of Earth's interior, which the author hopes could eventually be iterated toward inferring precursor information relevant to earthquakes — motivated by the genuine and well-recognized difficulty of directly measuring the stress and strain-energy state of the deep crust with any existing instrument. We treat this revision on its merits: replacing neutrino–neutrino 'collision' with a network of ordinary matter-based detectors capturing an attenuated transiting beam is a substantive improvement, because it maps onto the same physical principle already used in real neutrino Earth tomography (IceCube) rather than an experimentally unobserved process. We confirm the transit-time kinematics (~ 42.5 ms across Earth's diameter, grade A/B) and review the same established physics base as before — CEvNS, geoneutrino detection, and IceCube-style absorption/oscillation tomography. Three quantitative gaps remain, and we grade them explicitly. First, no known technology produces a directed, usable-intensity neutrino beam without a GeV-scale accelerator and long decay pipe; a lunar site solves none of this (grade E for the claim that vacuum/quiet alone suffices). Second, because Earth's diameter (12,742 km) is negligible next to the Earth–Moon distance (384,400 km), every near-Earth receiving station — regardless of its position around the planet — sits at essentially the same enormous distance from a lunar source; scaling the COHERENT experiment's 19.3 m baseline to this distance dilutes event rate by roughly 4×10^{14} for a fixed source and detector (grade D, illustrative). Adding stations therefore improves angular coverage for tomographic reconstruction — a real and valuable idea, grade B/D by analogy with existing multi-baseline tomography — but does not by itself solve the sensitivity gap, which is identical at every station. Third, even IceCube's success with a cubic-kilometre detector and a strong natural neutrino flux required roughly a decade of accumulated data to resolve a handful of concentric global density shells; 'real-time' (day-to-week) resolution of localized crustal anomalies is therefore many further orders of magnitude beyond what any known neutrino technique can currently deliver (grade E for a real-time claim). Finally, even granting a future solution to all of the above, the specific hypothesis that crustal density anomalies precede earthquakes in a detectable, generalizable way has already been tested for fifty years under its classical name — the dilatancy–diffusion hypothesis — using instruments (seismic velocity, radon, GPS strain) far more sensitive at the relevant spatial scale than any neutrino technique could plausibly become; the consensus finding is that laboratory-observed precursory dilatancy has not been reliably reproduced in field data, largely because natural tectonic strain

rates (10^{-17} – 10^{-9} s⁻¹) are eight to twelve orders of magnitude slower than laboratory strain rates, a scale gap that a new detection modality does not by itself resolve. We conclude that the revised, capture-network framing is a genuine scientific improvement over the original collision concept and deserves to be developed along those lines — as a contribution to neutrino Earth tomography — while the earthquake-prediction application remains, honestly, an evidence-grade-E aspiration rather than a specified research pathway.

Keywords

Neutrino physics; neutrino Earth tomography; coherent elastic neutrino–nucleus scattering; geoneutrinos; dilatancy–diffusion hypothesis; earthquake prediction; conceptual critique; evidence grading

1. Introduction and Motivation

This note revises an earlier critical assessment (v0.1) of a proposal originally framed as a 'Moon–orbital neutrino wall collision experiment.' Following that assessment, the author refined the concept: rather than causing two neutrino beams to 'collide,' the revised architecture places a neutrino source on the Moon and multiple receiving 'neutrino wall' stations at various points in near-Earth orbit. Each station would record the timing, attenuation, and any anomalous sub-threshold events in the neutrino flux that has just transited the Earth, with the resulting multi-station data set used for real-time monitoring of the planet's interior. The author's stated motivation is direct and, on its own terms, well-founded: earthquakes result from extreme stress building up in the crust and lithosphere under plate motion (ultimately driven by deeper processes including inner-core dynamics), and no existing instrument can directly and precisely measure the stress or kinetic energy acting on the deep crust. The hope is that a sufficiently rich, real-time neutrino-transit data set could, through iterative refinement, be developed into a tool for inferring that otherwise-unmeasurable state, and eventually for earthquake detection and prediction.

We again apply the evidence-grading discipline used throughout the author's LX-RSRS and LX-NWT v0.1 work. The headline finding of this revision is genuinely positive: moving from a neutrino–neutrino collision mechanism (which has no experimental precedent at any energy) to a multi-station capture network (which measures neutrino–matter interactions in ordinary detector material, the same physical process already used in real Earth tomography) removes the single largest, unphysical element of the original proposal. What remains are large but different-in-kind problems — detector sensitivity, integration time, and the still-unresolved question of whether crustal density anomalies are a reliable earthquake precursor at all — each of which we assess in turn.

2. Established Physics Background

2.1 Neutrino Transit Kinematics

Using Earth's mean diameter of 12,742 km (1.2742×10^7 m) and $c = 299,792,458$ m/s, a particle travelling at c crosses the full diameter in $t \approx 0.0425$ s (≈ 42.5 ms). This is correct and uncontroversial (grade A/B); it establishes only a timing constant and says nothing about whether a given architecture can produce, direct, or detect neutrinos at a useful rate, which is the subject of Sections 3–5.

2.2 Neutrino Interaction Cross Sections

Neutrinos interact only via the weak force, so cross sections with ordinary matter are extremely small. Inverse-beta-decay on free protons is of order 10^{-43} – 10^{-44} cm² (grade A). Coherent elastic neutrino–nucleus scattering (CEvNS), first observed by COHERENT in 2017, has a larger cross section — roughly $(1.6\text{--}6) \times 10^{-39}$ cm² for MeV-scale neutrinos on medium-to-heavy nuclei, scaling approximately with the square of the neutron number —

which is why it could be seen with a comparatively small (kilogram-scale) detector at short range (19.3 m from an intense accelerator-driven source). It took 44 years from theoretical prediction to first detection (grade A/B). This remains the most relevant real precedent for any 'wall'-style detector envisioned in the proposal, since each receiving station is, physically, a neutrino–matter (not neutrino–neutrino) detector.

2.3 How Directed Neutrino Beams Are Produced

Neutrinos cannot be focused or steered after production; every existing directional 'neutrino beam' (T2K, NuMI) gets its directionality entirely from the kinematics of the production step — a GeV-scale proton beam striking a target to produce pions and kaons, which are magnetically focused while still charged and then decay in flight inside a helium-filled pipe 100–700 m long, with the forward boost of the parent hadron carrying over to the neutrino. A 'high-vacuum, low-noise' site addresses only detection-side background, not production; the source material identifies no mechanism by which a lunar site substitutes for the accelerator/decay-pipe chain (grade C for the existing accelerator technology; grade E for the claim that a lunar site removes the requirement for it).

2.4 Real, Active Neutrino Earth-Probing Techniques

Geoneutrino experiments (KamLAND, Borexino, joined more recently by SNO+ and JUNO) detect antineutrinos from natural radioactive decay in the Earth using deep-underground, kiloton-scale liquid-scintillator detectors, currently favouring roughly 20 TW of present-day radiogenic heating (grade A/B). Neutrino Earth tomography uses naturally occurring TeV-scale atmospheric neutrinos together with a cubic-kilometre detector (IceCube) to measure flux attenuation as a function of path length and angle through the Earth, from which the Earth's total mass, core mass, moment of inertia, and (via oscillation) a layered-versus-homogeneous density structure have been inferred, using roughly a decade of accumulated data (grade A/B, results published through 2025–2026). These are the closest existing analogues to the revised proposal: both use natural, high-statistics neutrino fluxes and very large detector masses accumulated over long exposure times — the two ingredients the revised architecture will need to substitute for, at planetary source-detector distances, with an artificial source and station-scale detectors.

3. What the Revised Proposal Adopts from Established Physics, and What It Does Not

The table below updates the honest-novelty comparison from v0.1 to reflect the shift from a collision mechanism to a capture network.

Element of the proposal	Status in the literature	Evidence grade
Neutrinos travel at $\sim c$; transit time across Earth's diameter ~ 42.5 ms	Direct application of special relativity; not contested	A/B
Neutrino interaction with ordinary matter at each receiving station (CEvNS or similar)	Real, observed interaction channel (COHERENT, 2017); this is what a 'wall' detector would actually measure	A
Directed, high-flux neutrino beam from a lunar source	No known production mechanism without a GeV-scale accelerator + long decay pipe; a vacuum/low-noise site does not supply one	E (as claimed for a lunar station); C for the underlying accelerator technology itself
Multiple receiving stations at different orbital positions, for tomographic reconstruction	Directly analogous to how additional baselines/angles improve resolution in any tomography, including IceCube-style neutrino tomography	B/D — sound design principle, not yet demonstrated at this scale
'Real-time' (day-to-week) anomaly monitoring	Existing successful neutrino Earth tomography (IceCube) required ~ 10 years of accumulated natural flux to resolve a	E

	handful of global shells; no known path to real-time resolution at any spatial scale	
Crustal density/stress anomalies as an earthquake precursor, detectable by any instrument	Directly corresponds to the classical dilatancy–diffusion hypothesis (Nur, Scholz et al., 1970s); tested for 50 years with instruments far more locally sensitive than neutrino tomography could become; not reliably validated at crustal scale (see Section 5.3)	E
Application of the resulting data to earthquake detection and prediction	No published mechanism links neutrino-derived density/timing data to seismic precursors; short-term earthquake prediction itself has no validated method by any technique	E — no known supporting mechanism

Compared with v0.1, one full row (neutrino–neutrino collision) is removed because the revised architecture no longer requires it. This is a genuine improvement in the proposal's physical grounding, even though several large gaps remain.

4. Quantitative Assessment of the Capture-Network Architecture

4.1 The Production Problem (Unchanged)

As in Section 2.3, no known technology generates a directional, usable-intensity neutrino beam without a full accelerator/decay-pipe production chain. This remains unresolved in the revised proposal exactly as it was in v0.1 (grade E for a lunar-only solution).

4.2 Why Multiple Stations Around Earth Do Not Change the Distance Problem

A key quantitative fact bears directly on the multi-station design: Earth's diameter (12,742 km) is only about 3.3% of the mean Earth–Moon distance (384,400 km). Consequently, whether a receiving station sits on the far side of the Earth from the Moon, on the near side, or anywhere else in near-Earth orbit (typical altitudes of hundreds to a few thousand km), its distance from a lunar source varies only by a few percent — utterly negligible next to the $\sim 4 \times 10^{14}$ dilution factor derived in Section 4.3 below. This means every station in the proposed network faces essentially the same sensitivity gap; the network improves the number and geometry of vantage points for reconstructing a spatial pattern (a real and useful tomographic idea, grade B/D), but it does not multiply the number of detected events or otherwise ease the core detection-sensitivity problem, which must be solved once, for a single representative baseline, before it is worth solving for many.

4.3 A Fermi Illustration: Scaling a Real Detection Baseline to Lunar Distance

As in v0.1, we take COHERENT's real CEvNS geometry — a 14.6 kg CsI detector 19.3 m from an intense accelerator-driven neutrino source — and ask what happens if that same source-detector pair, with identical source intensity and detector mass, were separated by 384,400 km instead. Flux (and event rate, for fixed cross section and mass) falls as $1/r^2$ absent absorption. The distance ratio is $3.844 \times 10^8 / 19.3 \approx 1.99 \times 10^7$, giving a dilution factor of about $(1.99 \times 10^7)^2 \approx 3.97 \times 10^{14}$ (grade D, illustrative order-of-magnitude only; it ignores beam collimation and any future increase in source intensity or detector mass, and is not a substitute for a full flux-and-sensitivity study). As noted in Section 4.2, this figure applies essentially unchanged to every station in the proposed network, since near-Earth orbital altitude is a negligible correction to the lunar baseline.

4.4 The 'Real-Time' Problem

A separate, additional gap concerns integration time. IceCube's Earth-tomography results — arguably the most successful real-world demonstration of using neutrinos to probe planetary structure — used roughly a decade of accumulated atmospheric-neutrino data, from a cubic-kilometre detector exposed to a strong, continuous natural flux, to resolve the Earth into a handful of concentric density shells (core, outer core, mantle). 'Real-time' monitoring, in the sense of detecting a localized anomaly over a day-to-week timescale relevant to earthquake precursors, is a categorically harder problem than the multi-year, global-shell reconstruction IceCube already found difficult; nothing in the current literature suggests a known path to closing this additional gap, independent of the distance/sensitivity gap in Sections 4.2–4.3 (grade E for a real-time claim on any timescale currently specifiable).

5. The Earthquake-Detection and Prediction Extension

5.1 The State of the Art in Earthquake Prediction

Short-term earthquake prediction remains, per the International Commission on Earthquake Forecasting for Civil Protection (2009), a problem for which 'the search for precursors that are diagnostic of an impending earthquake has not yet produced a successful short-term prediction scheme,' despite decades of investigation across electromagnetic, hydrological, gas-geochemical, geodetic, and seismic precursor candidates (grade A: consensus position of the field).

5.2 What Neutrino Tomography Currently Measures

Geoneutrino detection measures an essentially static, spatially unresolved quantity (integrated radiogenic heat over years of exposure, dominated by lithosphere within ~500 km of the detector). IceCube-style tomography currently resolves a handful of concentric global shells over roughly a decade of data. Neither technique, in its current or foreseeable near-term form, is a candidate for resolving the localized, time-varying density or stress changes in a specific shallow fault zone that would be relevant to an individual earthquake (grade A/B for what these techniques do measure; grade E for the extrapolation to fault-zone monitoring).

5.3 The Closest Real Analogue: The Dilatancy–Diffusion Hypothesis

The proposal's underlying physical hypothesis — that crustal stress produces a detectable density or volume anomaly ahead of rupture — is not new; it is essentially the classical dilatancy–diffusion hypothesis developed by Nur, Scholz, and collaborators in the early 1970s, which holds that rising stress causes microcracking and volume increase (dilatancy) in crustal rock, with associated changes in seismic wave velocity, electrical resistivity, and pore-fluid pressure. This hypothesis was taken seriously enough to motivate the widely cited (and contested) 1975 Haicheng earthquake prediction, and has since been tested for roughly fifty years using instruments — borehole seismometers, GPS/InSAR strain measurement, radon monitoring, resistivity surveys — that are far more locally sensitive, at the relevant spatial scale of a single fault zone, than any neutrino-based technique could plausibly become given the sensitivity gaps in Section 4. The consensus finding of recent critical reviews (e.g., a 2025 retrospective marking the 50th anniversary of Haicheng) is that laboratory-observed dilatancy-related precursors have not been reliably reproduced in field data, and that a central reason is a stark scale gap: natural inter-seismic crustal strain rates are on the order of 10^{-17} to 10^{-9} per second, versus roughly 10^{-5} per second in laboratory rock-deformation experiments — a difference of eight to twelve orders of magnitude that field observations at any single site have not been able to bridge. This matters directly for the neutrino-wall proposal: even if the detection-sensitivity and real-time problems of Section 4 were somehow solved, the specific claim that crustal density anomalies reliably and generalizably precede earthquakes remains an open, actively debated question after fifty

years of study with dedicated, far more locally sensitive instruments — it is not a settled physical fact that a new detection modality could simply confirm (grade E, pre-existing open problem in geophysics, not created or resolved by this proposal).

5.4 Absence of a Known Mechanism Linking Neutrino Data to Seismic Precursors

Combining Sections 5.2 and 5.3: this note's literature search did not locate any published mechanism connecting neutrino interaction rates, of any kind, to pre-seismic geophysical changes, and the closest analogous non-neutrino technique (dilatancy monitoring) has itself not achieved a validated predictive result after five decades. We record the earthquake-prediction application explicitly as an evidence-grade-E hypothesis resting on two independently unresolved problems — neutrino detection at the required sensitivity, and the underlying seismological precursor hypothesis — rather than attempting to construct a plausible-sounding physical story connecting them.

6. More Defensible Directions Worth Pursuing

The author's underlying interest — using neutrinos, which pass through the Earth largely unimpeded, to probe otherwise inaccessible interior states — remains scientifically sound, and the shift to a capture-network framing moves the proposal closer to genuinely open, active research questions rather than further away from them:

- Neutrino Earth-tomography resolution: contributing to the statistical/instrumental methods needed to move IceCube-style tomography toward finer spatial resolution is an explicitly stated goal of the current IceCube Upgrade programme (grade B/C, active engineering problem).
- Multi-baseline tomographic reconstruction: the idea of using several simultaneous vantage points to improve spatial resolution is sound tomography design and worth developing formally (e.g., as a simulation study of achievable resolution versus number/geometry of stations, using currently known neutrino fluxes and cross sections) even before any new source technology exists (grade D, well-posed research question).
- Geoneutrino site geology: refining local crustal radioisotope models used to interpret existing KamLAND/Borexino data, a leading source of interpretive uncertainty the geoneutrino community has explicitly flagged (grade A/B, open problem).
- Multi-messenger crustal-stress monitoring: combining existing, locally sensitive precursor candidates (seismic velocity changes, radon, GPS/InSAR strain, resistivity) is the active frontier of dilatancy-related earthquake research, and does not require solving any neutrino-detection problem first (grade E as a predictive method, but a well-posed, actively studied research question).

7. Limitations of This Assessment

- This note is a critical, literature-grounded assessment, not an exhaustive review; a systematic search may locate additional relevant proposals or techniques not covered here.
- The Fermi illustration in Section 4.3 is a simplified geometric scaling argument, not a full neutrino-beam and detector-sensitivity calculation; it does not specify a source intensity, beam divergence, or detector technology, none of which the proposal specifies either.
- This note does not attempt to prove a universal negative; it reports the absence of any currently known or published supporting mechanism, the relevant standard for an evidence-grade-E classification, and remains open to revision if new theoretical or experimental work changes this picture.
- Numerical values are drawn from the cited literature at the precision reported there; the author should independently verify each figure, author list, and citation detail before any formal submission.

8. Conclusion

This revision credits a genuine improvement in the author's proposal: replacing an experimentally unobserved neutrino–neutrino 'collision' mechanism with a multi-station network of ordinary matter-based detectors capturing an attenuated, transiting neutrino beam brings the architecture's underlying physics into the same category as real, active neutrino Earth-tomography research (IceCube, geoneutrinos), rather than requiring new fundamental physics. What remains are large, quantifiable gaps rather than an unphysical mechanism: a lunar directional-source production problem (Section 4.1), a detection-sensitivity problem that is essentially identical at every station in the network because near-Earth orbital altitude is negligible next to the Earth–Moon distance (Sections 4.2–4.3), an integration-time problem that makes 'real-time' monitoring categorically harder than the multi-year global reconstructions IceCube has achieved (Section 4.4), and — independently of all of the above — a fifty-year-old, still-unresolved question in seismology itself about whether crustal density/volume anomalies reliably precede earthquakes at all (Section 5.3). We recommend that the author's next step be a quantitative simulation study of the multi-station tomographic-resolution idea using known neutrino fluxes and cross sections (Section 6), clearly separated from the earthquake-prediction aspiration, which should continue to be labelled, honestly, as an evidence-grade-E hypothesis awaiting solutions to problems that are currently open in two different fields.

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NASA JPL Horizons System / IAU reference value — mean Earth–Moon distance, 384,400 km.

Note: several reference entries above were located as secondary leads during AI-assisted literature search. Before any formal submission, the author should independently verify each entry's complete author list, volume/issue/page numbers, and DOI.